

Algebra 1
Unit 2
Lesson 7 Practice

Name Answer Key
Date _____
Period _____

1. Find the product of $(x^2 + 4)(2x - 5)$ using the distributive property.

$$2x^3 - 5x^2 + 8x - 20$$

2. Find the product of $(x^2 + 2)(3x + 4)$ using an area model.

	x^2	$+2$
$3x$	$3x^3$	$6x$
$+4$	$4x^2$	8

$$3x^3 + 4x^2 + 6x + 8$$

3. Find the product of $(-\frac{1}{2}y + 4y^2 - 3)(4y - 5)$ using the distributive property.

$$16y^3 - 22y^2 - \frac{19}{2}y + 15$$

4. Find the product of $(\frac{1}{3}y + 2y^2 - 6)(y - 4)$ using an area model.

	$\frac{1}{3}y$	$2y^2$	-6
y	$\frac{1}{3}y^2$	$2y^3$	$-6y$
-4	$-\frac{4}{3}y$	$-8y^2$	24

$$2y^3 - \frac{23}{3}y^2 - \frac{22}{3}y + 24$$

Find each product.

5. $(-4a + \frac{3}{5}b)^2$

$$16a^2 - \frac{24}{5}ab + \frac{9}{25}b^2$$

6. $(2.2 - 5c)(0.7 + 4c^3 - c^2)$

$$-20c^4 + 13.8c^3 - 2.2c^2 - 3.5c + 1.54$$

7. $(\frac{3}{5}x - 2)(\frac{-1}{2} + 4x)$

$$\frac{12}{5}x^2 - \frac{83}{10}x + 1$$

8. $(-4\frac{1}{2} + 4d^2)(5d - 7.1c - 2d^2)$

$$-22.5d + 31.95c + 9d^2 + 20d^3 - 28.4cd^2 - 8d^4$$

9. $(14 + 4m)(-6n - \frac{14}{5})$

$$-84n - 24mn - \frac{196}{5} - \frac{56}{5}m$$

10. $(3r^2 - \frac{5}{4} + 15r)(-2r + 3r^2 + 12)$

$$9r^4 + 39r^3 + \frac{9}{4}r^2 + \frac{365}{2}r - 15$$